

Capability-Specific Degradation Patterns in Quantized Small Language Models

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Abstract

Post-training quantization is the default way to deploy small language models (1–4B parameters), yet its impact is usually summarized by a single aggregate accuracy number that can hide severe failures in individual capabilities. We evaluate seven open instruction-tuned small LLMs spanning five architecture families (Qwen2.5, Llama-3.2, Gemma-2, Phi-3.5, and SmolLM2) at FP16 and 4-bit (bit-sandbytes NF4) across six capabilities: factual knowledge, commonsense, mathematical reasoning, multilingual math, code generation, and instruction following. Rather than reporting only mean accuracy, we construct a *capability-level degradation map* and ask whether capabilities degrade in a consistent order across architectures. We find that degradation is strongly capability-specific: multilingual mathematical reasoning and code generation are the most fragile capabilities (relative drops up to 57%), while commonsense reasoning is almost perfectly preserved ($\sim 2\%$) everywhere. However, the *ordering* of degradation is only weakly consistent across architectures (mean pairwise Kendall’s $\tau = 0.29$): the most- and least-fragile capabilities are shared, but the middle ordering is architecture-dependent. We further show that aggregate scores mask these failures—several models lose more than a fifth of a specific capability while their mean barely moves—and we document a measurement pitfall in which a heavily chat-tuned model scores near zero on code under raw-completion prompting yet degrades by 57% once prompted correctly. Our results argue that small-model quantization should be reported per capability, not in aggregate.

1 Introduction

Small language models in the 1–4B parameter range are increasingly the deployment-relevant artifact: they run on a single consumer GPU, on laptops, and on edge devices. Post-training

quantization—most commonly to 4 bits—is the standard way to fit these models into tight memory budgets. The practical question for a practitioner is simple: *what do I lose when I quantize?*

The usual answer is an aggregate benchmark score, or an average over a benchmark suite. We argue this is the wrong granularity. A model whose average accuracy drops by three points may have lost almost nothing on commonsense reasoning while losing a quarter of its code-generation or multilingual ability. For a deployed system that distinction is the whole story: an aggregate number that looks acceptable can conceal a capability that has effectively broken.

We therefore study quantization at the level of individual capabilities. We evaluate seven instruction-tuned small models from five architecture families at FP16 and 4-bit across six capabilities, and we organize the results as a *degradation map*: for every model and capability, the relative drop from FP16 to 4-bit. We then ask whether different capabilities degrade in a *consistent order* across architectures—if they do, the pattern is universal and predictable; if they do not, degradation is architecture-dependent and must be measured per model.

Our contributions are:

1. A capability-level degradation map for seven open small LLMs (five families) under 4-bit quantization across six capabilities, using a single controlled evaluation pipeline (Section 3).
2. Evidence that degradation is strongly *capability-specific*: multilingual math and code degrade most, commonsense least, with more than a $25\times$ spread in relative degradation within a single model (Section 4.1).
3. A cross-architecture consistency analysis (mean Kendall’s $\tau = 0.29$) showing that

the degradation *ordering* is only weakly universal: the extremes are shared but the middle ranking is architecture-dependent (Section 4.2).

4. A size-axis analysis showing that smaller models degrade more in magnitude, and a demonstration that aggregate scores hide capability-specific collapse, including a prompting-protocol pitfall on code evaluation (Sections 4.3–4.4).

2 Related Work

Unified compression benchmarks. UniComp (Author(s) to confirm, 2026b) compares pruning, quantization, and distillation across 40+ datasets and reports a “knowledge bias”: knowledge-style tasks are preserved while reasoning, multilingual, and instruction-following abilities degrade. Their focus is *comparing methods* on largely mid-to-large models; they do not isolate the small-model regime or produce an explicit per-capability degradation ranking, which is our focus.

Quantization and reasoning. Author(s) to confirm (2025b) provide a math-specific analysis, reporting up to 32% degradation on Llama-3 mathematical reasoning under AWQ/GPTQ. We corroborate the fragility of mathematical reasoning but situate it among five other capabilities and across architectures and model sizes.

Multilingual quantization. Marchisio et al. (2024) show that quantization disproportionately harms non-Latin-script languages and that automatic metrics can underestimate the damage. Consistent with this, we find multilingual math (MGSM) to be the single most fragile capability in our suite.

Coding and STEM. A recent quantization benchmark (Author(s) to confirm, 2025a) finds coding and STEM tasks degrade most on MT-Bench, but is Llama-centric and oriented toward comparing quantization methods rather than ranking capabilities across architectures.

Mechanisms. Author(s) to confirm (2026a) give a mechanistic account of why extreme (2-bit) quantization fails, identifying distinct failure modes. Our study is complementary: we measure benchmark-level capability degradation at the

deployment-relevant 4-bit setting rather than analyzing internal mechanisms.

Positioning. Relative to this work we (i) restrict to the understudied 1–4B small-model regime, (ii) compare five architecture families under one identical pipeline, (iii) report an explicit capability degradation *ordering* and test its cross-architecture consistency, and (iv) add a size axis within two families.

3 Experimental Setup

Models. We evaluate seven instruction-tuned models. Five form the cross-architecture set used for the consistency analysis: Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, Gemma-2-2B-it, Phi-3.5-mini-instruct (3.8B), and SmolLM2-1.7B-Instruct. Two additional models form a size axis within existing families: Qwen2.5-1.5B-Instruct and Llama-3.2-1B-Instruct. We use instruction-tuned variants because instruction following (IFEval) requires them and because they are the deployment-relevant artifact at this scale. All models are ≤ 4 B parameters by design; Gemma-2 is loaded with eager attention so its logit soft-capping is applied correctly.

Quantization. We compare two precisions: FP16 (baseline) and 4-bit weight quantization via bitsandbytes NF4 (Dettmers et al., 2022). We focus on the aggressive 4-bit regime because it is where capability-specific failures emerge; 8-bit quantization was near-lossless in pilot runs and is omitted. We note GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) as alternative 4-bit methods; we standardize on NF4 to hold the quantization method fixed across all architectures, isolating architecture as the variable of interest.

Capabilities and benchmarks. We map six capabilities to standard benchmarks, evaluated with the LM Evaluation Harness (Gao et al., 2024): factual knowledge (ARC-Challenge (Clark et al., 2018), 25-shot, acc_norm); commonsense (HellaSwag (Zellers et al., 2019), 10-shot, acc_norm); mathematical reasoning (GSM8K (Cobbe et al., 2021), 8-shot, flexible-extract exact match); multilingual math (MGSM (Shi et al., 2022), 8-shot, exact match, macro-averaged over the standard languages); code generation (HumanEval (Chen et al., 2021), 0-shot, pass@1); and instruction following (IFEval (Zhou et al., 2023), 0-shot, prompt-level strict accuracy). Generative tasks

(GSM8K, MGSM, IFEval) use the chat template; HumanEval uses raw completion (the standard protocol); multiple-choice tasks use loglikelihood scoring. MGSM uses its standard language set and does not include Azerbaijani or Turkish.

Evaluation pipeline. All runs use a single controlled pipeline. Generative tasks are executed with a high-throughput vLLM backend; multiple-choice loglikelihood tasks use the Hugging Face Transformers backend. The same backend is used for a given capability across all precisions and models, so any FP16-vs-4-bit difference within a capability reflects quantization and not the harness. We verified pipeline correctness by matching FP16 baselines against published numbers (e.g., Qwen2.5-3B GSM8K and ARC). All $7 \times 2 \times 6 = 84$ runs completed.

Analysis. For each model and capability we compute the relative degradation

$$\Delta_{\text{rel}} = \frac{s_{\text{FP16}} - s_{\text{4bit}}}{s_{\text{FP16}}} \times 100\%.$$

To avoid unstable ratios, we exclude any capability whose FP16 baseline is below 10 points from the relative-degradation analysis (these are reported as absolute scores) and additionally report absolute drops as a sanity check. We rank capabilities by Δ_{rel} within each model (rank 1 = most degraded) and compare rankings across models with Kendall’s τ (Kendall, 1938). A high mean τ indicates a universal degradation order; a low mean τ indicates architecture-dependence. We separately compute τ between the small and large member of each family (the size axis).

4 Results

4.1 Capability-Specific Degradation

Table 1 reports the relative degradation map and Table 2 the underlying absolute scores. The headline observation is the very large spread *within* a single model. For Qwen2.5-1.5B, relative degradation ranges from 2.6% (HellaSwag) to 22.9% (MGSM) and 22.6% (HumanEval)—an order-of-magnitude difference between the most- and least-affected capability. Aggregating these into one number would report a moderate drop and erase the fact that two capabilities lost roughly a quarter of their performance.

Two capabilities stand out as consistently fragile. **Multilingual math (MGSM)** is the most degraded capability for five of the six models for

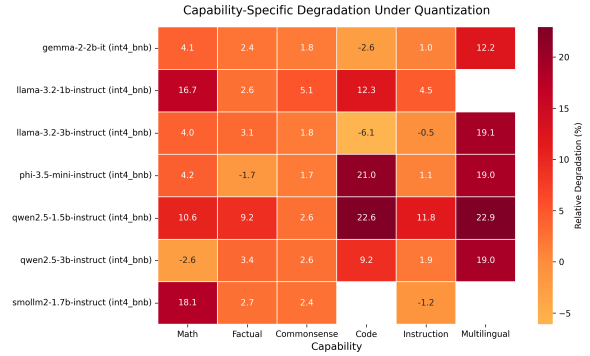


Figure 1: Relative degradation (%) per model (rows) and capability (columns). Multilingual math and code are the hot columns; commonsense is uniformly cool.

which it is measurable, with relative drops of 12–23%. **Code generation (HumanEval)** is the most volatile: catastrophic for some models (Qwen2.5-1.5B 22.6%, Phi-3.5 21.0%, and SmoLLM2 57%; see Section 4.4) yet statistically unchanged for others (Llama-3.2-3B, Gemma-2). At the other extreme, **commonsense (HellaSwag)** is essentially preserved everywhere (1.7–5.1%), and factual knowledge (ARC) is largely robust. Small negative values (4-bit marginally above FP16) occur for a few code/math cells and are within evaluation noise. Figure 1 visualizes the map.

4.2 Cross-Architecture Consistency

We rank the six capabilities by degradation within each model and compare rankings across all $\binom{7}{2} = 21$ model pairs with Kendall’s τ . The mean pairwise τ is **0.29**, which we interpret as *weak* consistency: the degradation order is *not* universal across architectures. The pattern is better described as partially consistent. The extremes are shared—multilingual math is reliably near the top of the ranking and commonsense reliably near the bottom—but the middle of the ranking (knowledge, math, code, instruction following) reorders from architecture to architecture, which depresses the mean τ . Pairwise agreement is correspondingly heterogeneous: some pairs rank capabilities identically (e.g., Gemma-2 vs. Llama-3.2-3B, $\tau = 1.0$) while others are uncorrelated (near $\tau = 0$). Figure 2 shows the rankings side by side.

The practical implication is that one cannot reliably predict a new architecture’s full degradation profile from another’s. A deployment decision should measure the specific model rather than assume a universal ordering. This is itself actionable: it argues against “rules of thumb” for which capa-

Model (4-bit vs FP16)	GSM8K	ARC	HellaSwag	HumanEval	IFEval	MGSM
Qwen2.5-3B	−2.6	3.4	2.6	9.2	1.9	19.0
Qwen2.5-1.5B	10.6	9.3	2.6	22.6	11.8	22.9
Llama-3.2-3B	4.0	3.1	1.8	−6.1	−0.5	19.1
Llama-3.2-1B	16.7	2.7	5.1	12.3	4.5	−
Gemma-2-2B	4.1	2.4	1.8	−2.6	1.0	12.2
Phi-3.5-mini	4.3	−1.7	1.7	21.0	1.1	19.0
SmolLM2-1.7B	18.1	2.7	2.4	57.2 [†]	−1.2	−

Table 1: Relative degradation Δ_{rel} (%) from FP16 to 4-bit (higher = worse). “−” marks capabilities excluded for an FP16 baseline below 10 points (MGSM for the two smallest models). [†]: SmolLM2 HumanEval under the chat-template protocol (Section 4.4); under the common raw-completion protocol its baseline is near zero and the cell is excluded.

Model	Prec.	GSM8K	ARC	HellaSwag	HumanEval	IFEval	MGSM
Qwen2.5-3B	FP16	63.4	60.7	75.3	53.1	59.3	24.9
	4-bit	65.1	58.6	73.3	48.2	58.2	20.2
Qwen2.5-1.5B	FP16	61.7	54.4	67.9	37.8	40.9	10.3
	4-bit	55.2	49.3	66.1	29.3	36.0	8.0
Llama-3.2-3B	FP16	77.3	52.8	73.6	50.0	69.3	31.0
	4-bit	74.2	51.2	72.3	53.1	69.7	25.1
Llama-3.2-1B	FP16	44.1	41.8	61.1	34.8	49.9	6.4
	4-bit	36.8	40.7	57.9	30.5	47.7	6.0
Gemma-2-2B	FP16	54.1	57.9	71.5	23.8	55.6	37.6
	4-bit	51.9	56.6	70.2	24.4	55.1	33.1
Phi-3.5-mini	FP16	78.6	64.8	79.0	61.0	49.9	33.0
	4-bit	75.3	65.9	77.7	48.2	49.4	26.8
SmolLM2-1.7B	FP16	49.1	53.2	72.6	34.2 [†]	48.2	3.9
	4-bit	40.3	51.8	70.8	14.6 [†]	48.8	3.2

Table 2: Absolute scores (%) per model and precision. [†]: SmolLM2 HumanEval under the chat-template protocol; under raw completion both values are near zero (see Section 4.4).

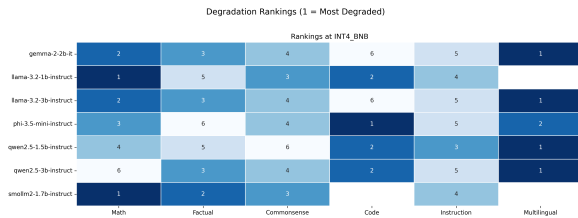


Figure 2: Degradation rankings per model (1 = most degraded). Agreement across columns would indicate a shared ordering; the visible reordering in the middle ranks reflects the low mean Kendall’s τ .

bilities survive quantization.

4.3 Size Axis

Within the two families evaluated at two sizes, the smaller model degrades *more in magnitude*. Qwen2.5-1.5B degrades more than Qwen2.5-3B on five of six capabilities (e.g., GSM8K 10.6% vs. −2.6%, HumanEval 22.6% vs. 9.2%); Llama-3.2-1B degrades more than Llama-3.2-3B on math and code (GSM8K 16.7% vs. 4.0%). The degradation *ordering*, however, does not strongly persist across sizes: the small-vs-large Kendall’s τ is 0.47

for Qwen2.5 and 0.00 for Llama-3.2. Quantization thus hurts smaller models more, but not in a fixed capability order even within a family.

4.4 Aggregate Scores Hide Capability Failures

The degradation map exposes failures that an aggregate would conceal. The clearest case is code generation under quantization for small chat-tuned models.

Under the standard raw-completion HumanEval protocol used for all models, SmolLM2-1.7B scores 0.6% at FP16 and 0.0% at 4-bit—numbers that, taken at face value, suggest the model simply cannot code. Inspection of its generations shows the cause is a *prompting-protocol artifact*: SmolLM2 is heavily chat-tuned and, given a bare function signature to continue, emits prose and markdown rather than code. The six other models continue raw code normally. When SmolLM2 is instead prompted with its chat template and the emitted function is extracted and executed, it scores 34.2% at FP16 and 14.6% at 4-bit—a **relative degradation of 57%, the largest code degra-**

dation in our study. The raw-completion protocol had hidden a near-total collapse of a real capability.

We draw two lessons. First, capability-specific evaluation can be sensitive to prompting protocol, and a single near-zero score should be audited before being read as inability. Second, and more importantly for our thesis, the underlying phenomenon is severe: a 1.7B model loses more than half of a genuine code-generation ability under 4-bit quantization, which no aggregate score in our suite surfaces. We report SmolLM2’s chat-protocol code result separately (the † entries in Tables 1 and 2) because it uses a different prompt format from the other models; the cross-model consistency analysis (Section 4.2) uses only the common protocol.

5 Discussion

Three themes emerge. **(1) Capability, not aggregate.** Across every model, quantization damage is concentrated in a minority of capabilities. Reporting a single accuracy number is not just lossy but actively misleading when the lost capability is the one a deployment depends on (e.g., multilingual or code). **(2) Fragility is partly intrinsic to the capability.** Multilingual math and code are fragile across architectures while commonsense is robust, suggesting the brittleness is tied to what the capability demands—long multi-step generation and exact-token correctness—more than to the architecture. **(3) But ordering is architecture-dependent.** The weak mean τ shows that beyond the shared extremes, which capability breaks next depends on the model, so practitioners cannot port a degradation profile across families. The deployment recommendation that follows is concrete: before shipping a quantized small model, re-test its multilingual and code capabilities specifically, and do not rely on an aggregate score or on another model’s profile.

Limitations

Our study covers seven models, one quantization method (NF4) at one bit-width (4-bit), and one benchmark per capability; broader coverage (AWQ/GPTQ, additional benchmarks per capability, more families) would strengthen the consistency estimates, whose per-pair p -values are limited by having only six capabilities. We report single evaluation runs; generative benchmarks

(GSM8K, HumanEval, MGSM) carry run-to-run variance, so small ranking differences should be read with the reported standard errors in mind. MGSM uses the standard language set and is excluded for the two smallest models, where the FP16 baseline is too low to form a stable ratio; degradation there is reported in absolute terms only. The Kendall’s- τ analysis ranks six capabilities, so individual pairwise correlations are noisy and we emphasize the mean. Finally, as Section 4.4 shows, code evaluation is protocol-sensitive for chat-tuned models; we mitigated this by auditing and re-prompting, but a fully uniform instruction-style protocol across all models is left to future work.

6 Conclusion

We presented a capability-level degradation map for seven small instruction-tuned LLMs under 4-bit quantization. Degradation is strongly capability-specific: multilingual math and code are most fragile (up to 57% relative loss) while commonsense is nearly untouched. The degradation *ordering* is only weakly consistent across architectures (mean Kendall’s $\tau = 0.29$), so the extremes are predictable but the full profile is architecture-dependent. Smaller models degrade more in magnitude. Because aggregate accuracy hides these capability-specific failures, we recommend that small-model quantization be reported and audited per capability.

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